[1] if ↻ x-axis ⇒ $V = \int_{y=c}^{y=d} 2\pi\,(\text{shell radius})(\text{shell height})\,dy$
or any line to it

and vice versa.

[3] height of cylinder
// Axis of rotation

[2] $x = d(\text{curve}, y\text{-axis})$
$y = d(\text{curve}, x\text{-axis})$

6.3

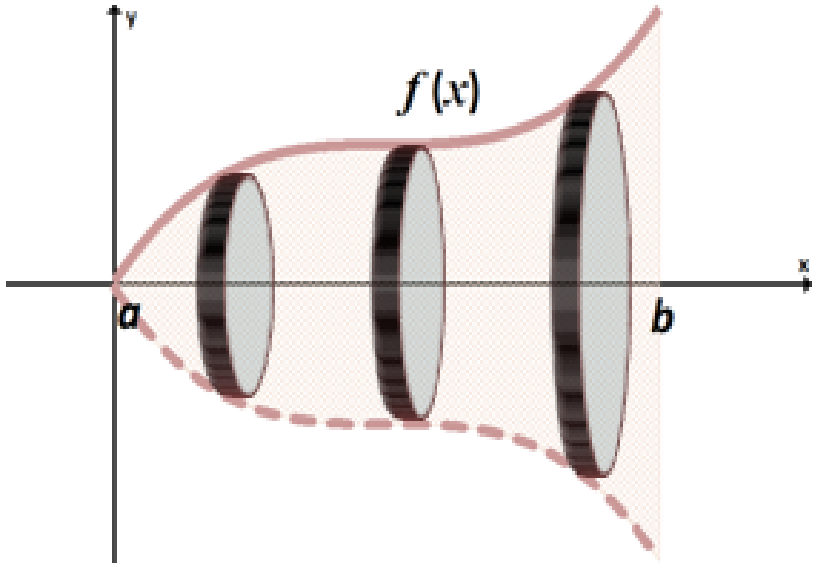
Volumes by Cylindrical Shells

$R = d(\text{curve}, \text{axis of rotation})$

Volumes of Solids: The Disk Method

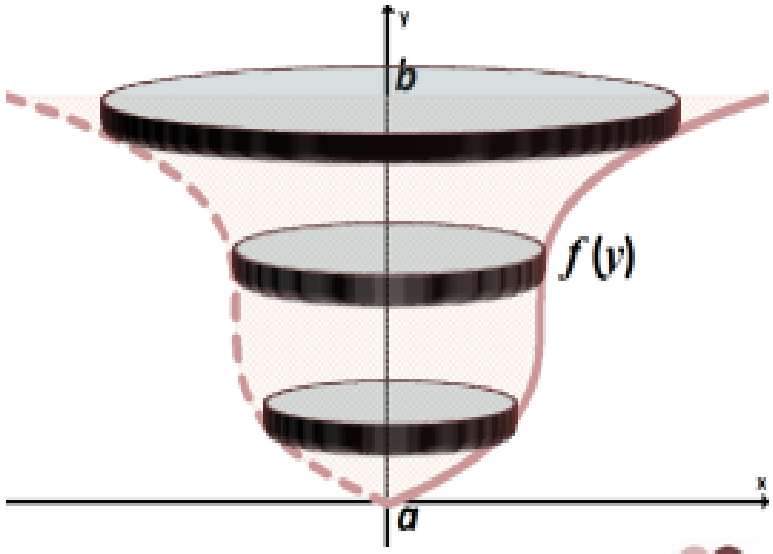
Horizontal Axis of Revolution:

$$\text{Volume} = \pi \int_a^b \left[f(x) \right]^2 dx$$



Vertical Axis of Revolution:

$$\text{Volume} = \pi \int_a^b \left[f(y) \right]^2 dy$$



She Loves Math

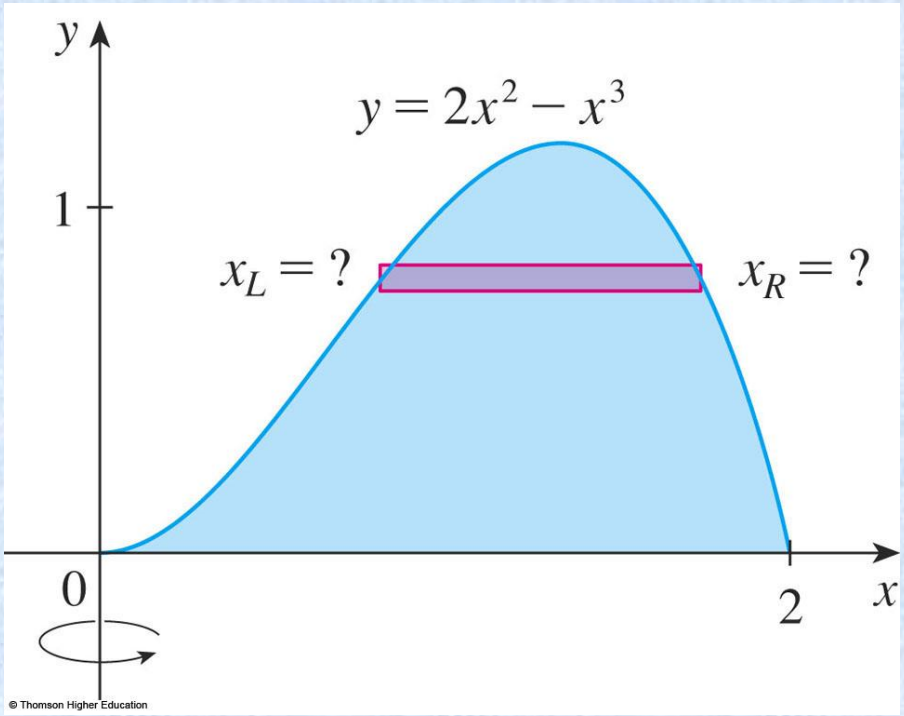
In this section, we will learn:
How to apply the method of cylindrical shells to find out the volume of a solid.

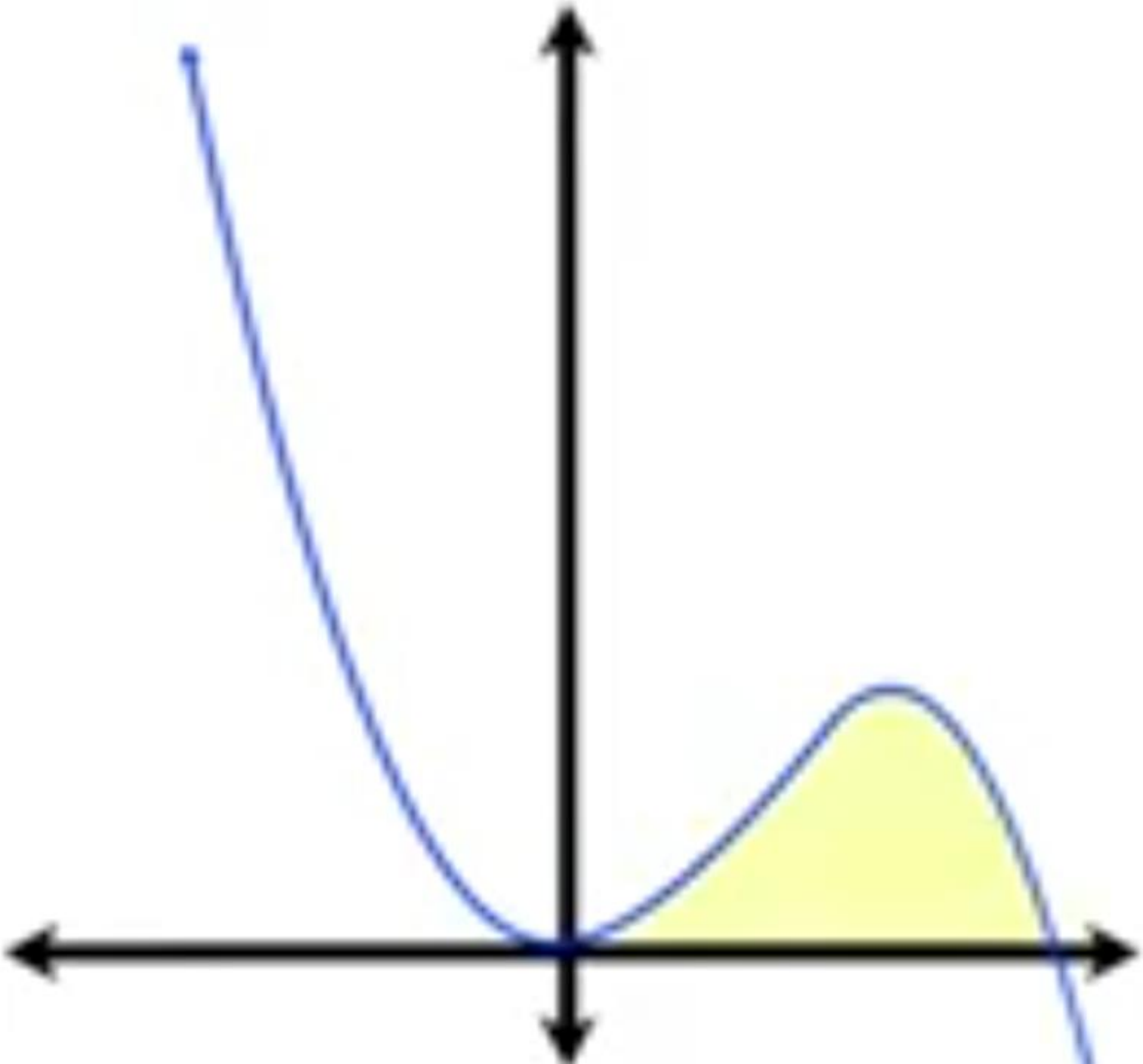
Some volume problems are very difficult to handle by the methods discussed in Section 6.2

Example: Let's consider the problem of finding the volume of the solid obtained by rotating about

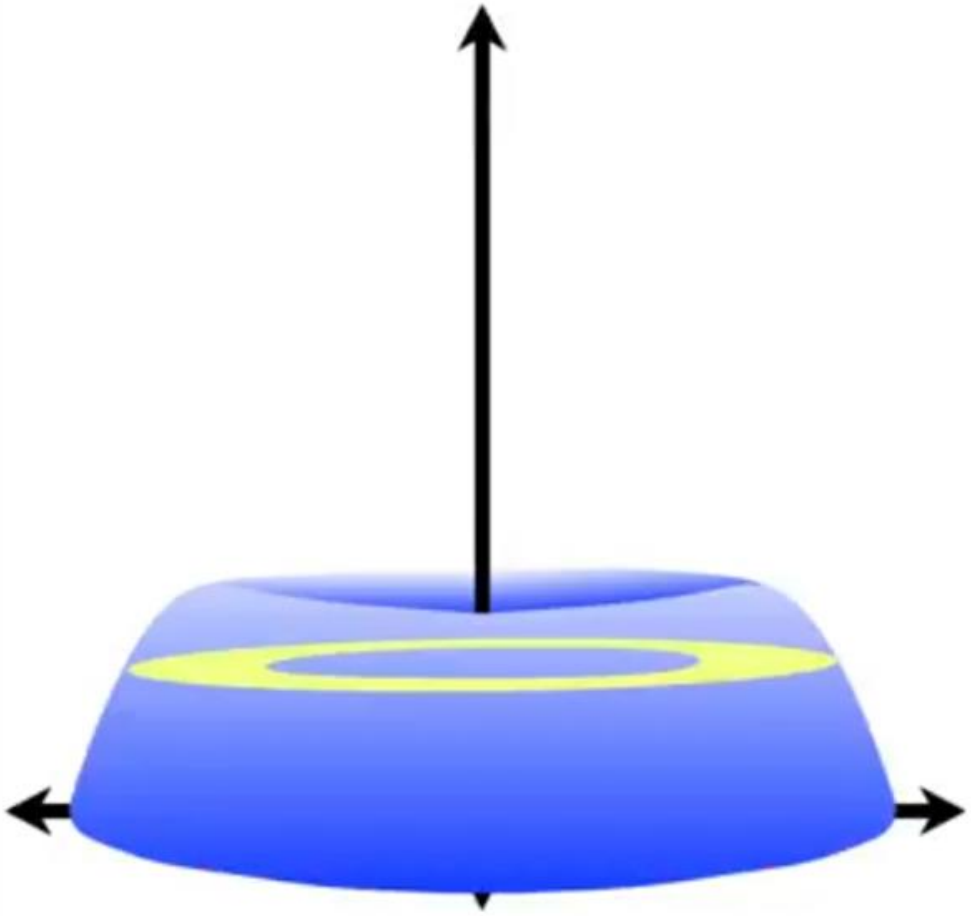
the y-axis the region bounded by $y = 2x^2 - x^3$

and $y = 0$.



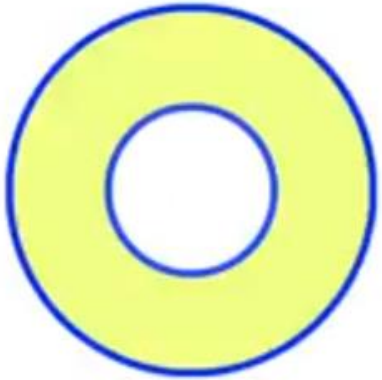


Calculating Volume by Cylindrical Shells



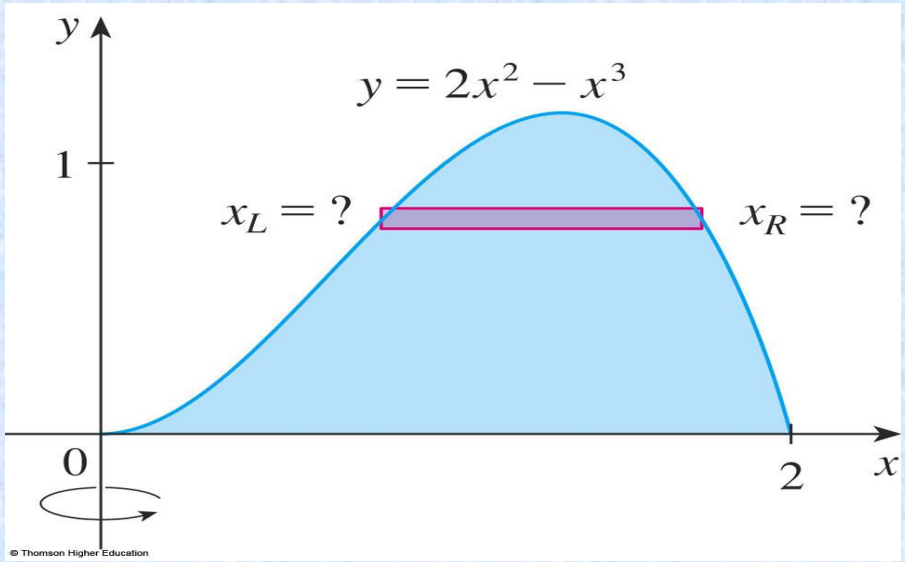
$y = 2x^2 - x^3$

washer cross section

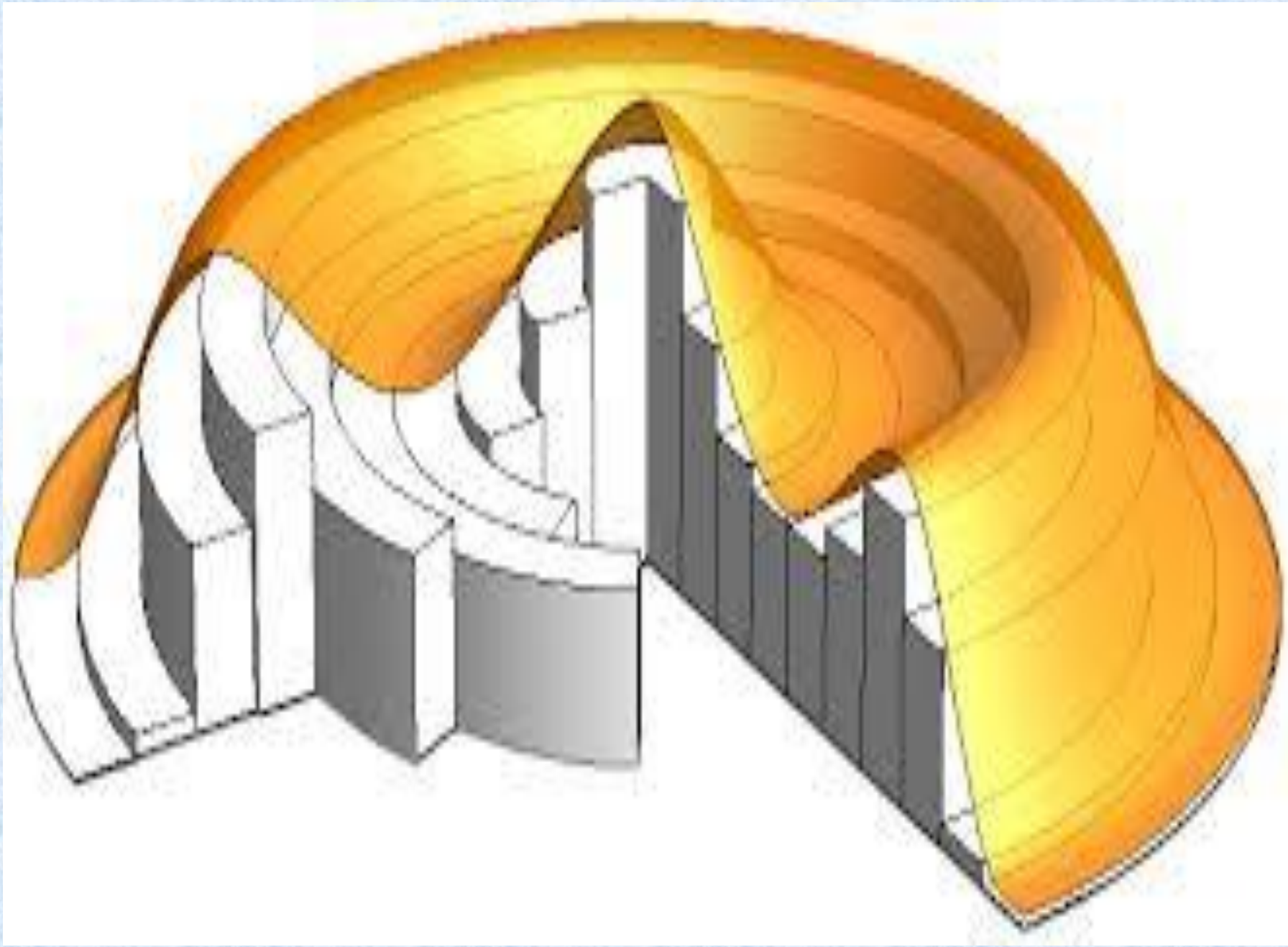


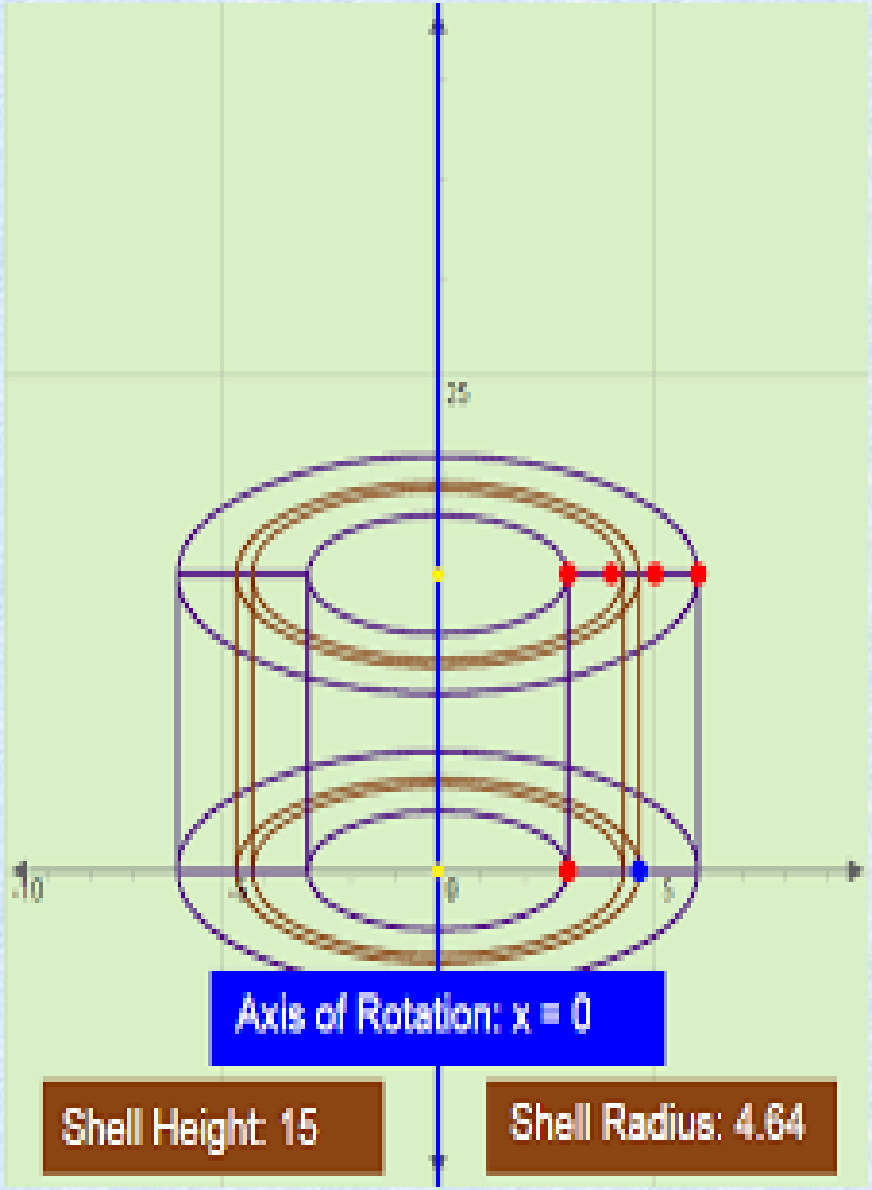
If we slice perpendicular to the *y*-axis, washer.

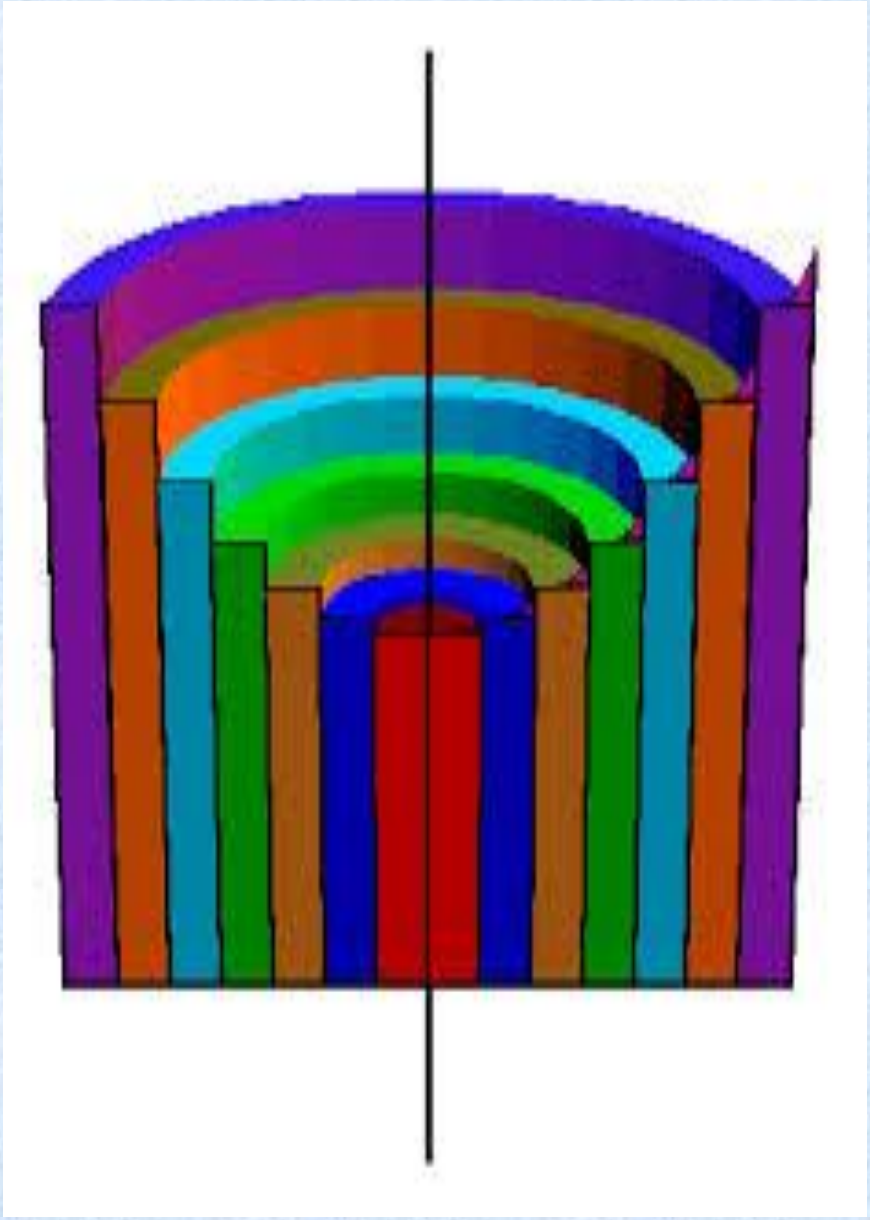
- However, to compute the inner radius and the outer radius of the washer, we would have to solve the cubic equation $y = 2x^2 - x^3$ for x in terms of y.



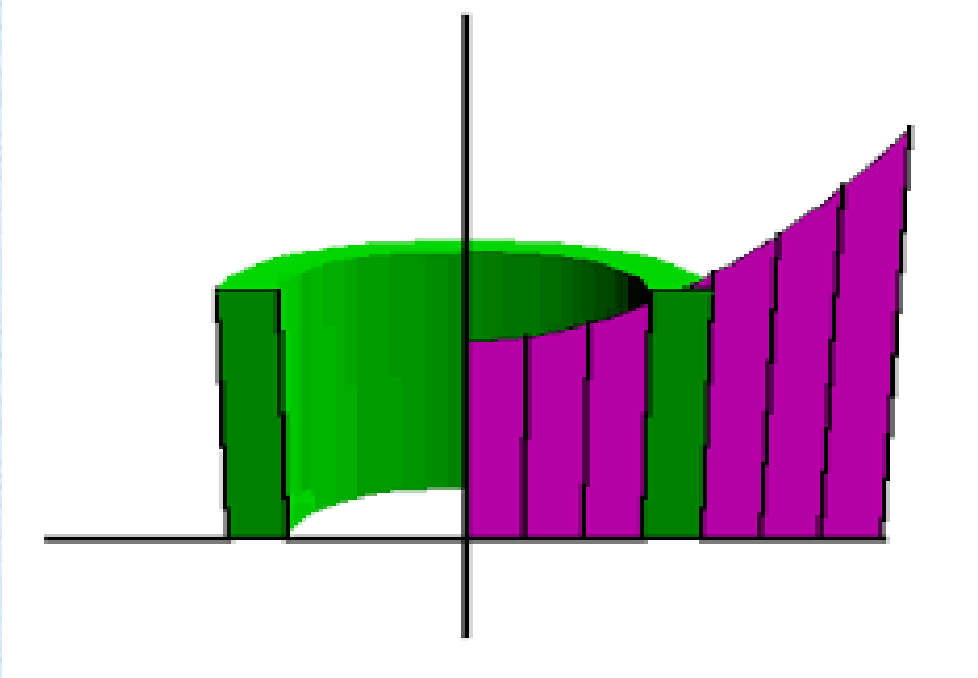
- That's not easy.




25
-10
-5
0
5
Axis of Rotation: x = 0
Shell Height: 15
Shell Radius: 4.64



Fortunately, there is a method (the method of cylindrical shells) that is easier to use in such a case.



If the axis of rotation is y- axis or parallel to it:

$$V = \int_a^b 2\pi x f(x)dx$$

Or

$$V = 2\pi \int_a^b (\textbf{shell radius})\,(\textbf{shell hight})\,\mathbf{dx}$$

1-The whole integral should be written in terms of x.

2- a and b are the limits of integration founded from x- axis.

3- draw the shell parallel to y-axis and hence the height will be the edge parallel to y-axis.

4-The radius of the shell is the distance between the shell and the axis of rotation.

5- Let x= the distance between y-axis and the shell.

Remark:

If the axis of rotation is y-axis or parallel to it and you have two curves f(x) and g(x) then:

$$V = 2\pi \int_a^b [f(x) - g(x)]\mathrm{dx}$$

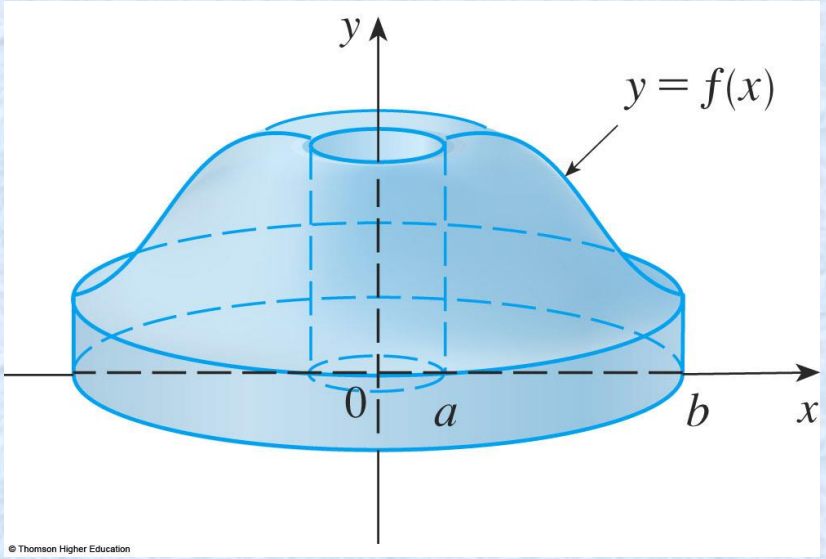
Where f is the top function and g is the bottom function.

Formula:

The volume of the solid obtained by rotating about the *y*-axis the region under the curve $y = f(x)$ from a to b, is:

$$V = \int_a^b 2\pi x f(x)\,dx$$

where $0 \le a < b$



Example 1

Find the volume of the solid obtained by rotating about the *y-axis* the region bounded by $y = -x^2 + x$ and $y = 0$.

Example 1

Find the volume of the solid obtained by rotating about the *line* $x=-3$ the region bounded by $y = -x^2 + x$ and $y = 0$.

Example

Find the volume of the solid obtained by rotating about the *y*-axis the region bounded by $y = \sin x^2$

Over $[0, \sqrt{\pi}]$

Example 1

Find the volume of the solid obtained by rotating about the $x = 2$ the region bounded by $y = 2x^2 - x^3$ and $y = 0$.

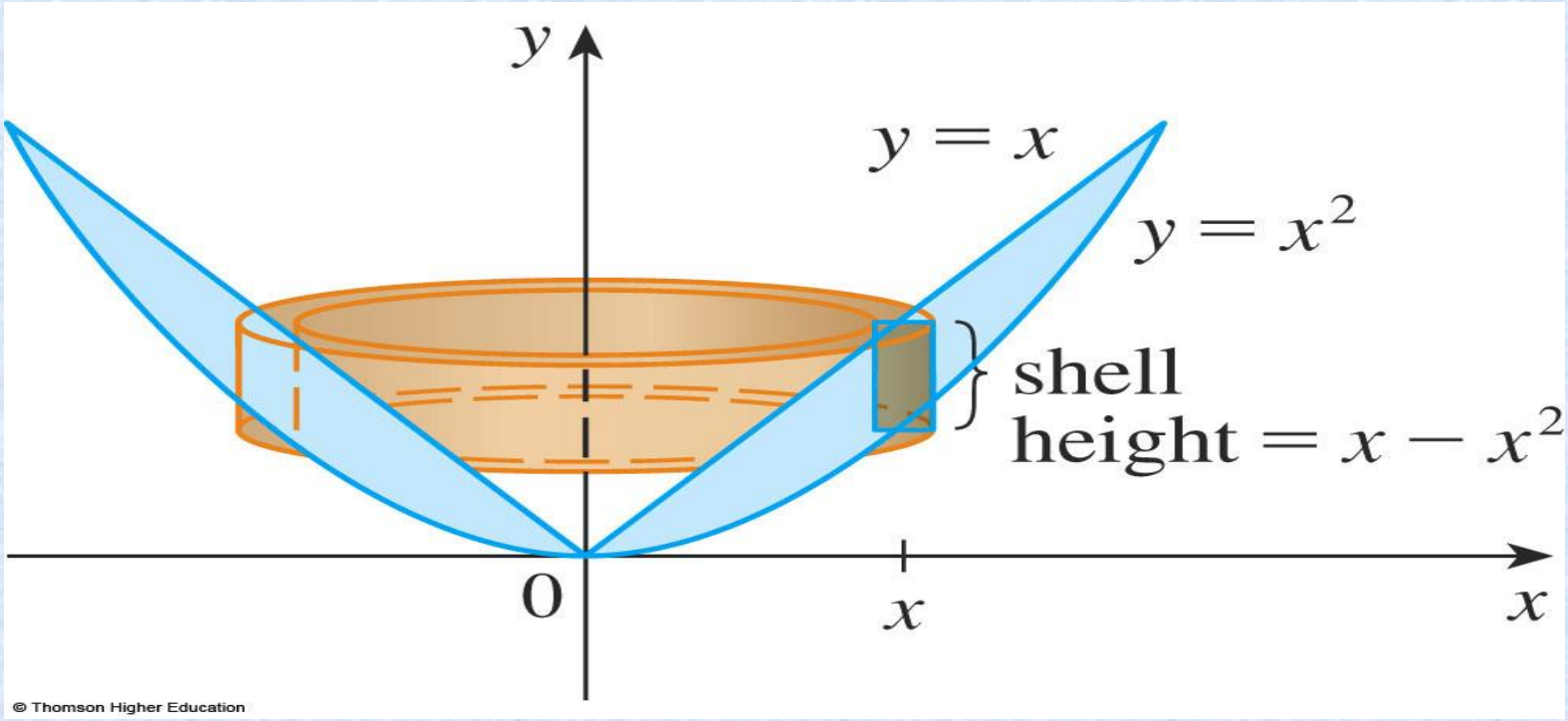
NOTE

However, in other examples, the methods learned in Section 6.2 may be easier.

Example 2

Find the volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$.

Example 2

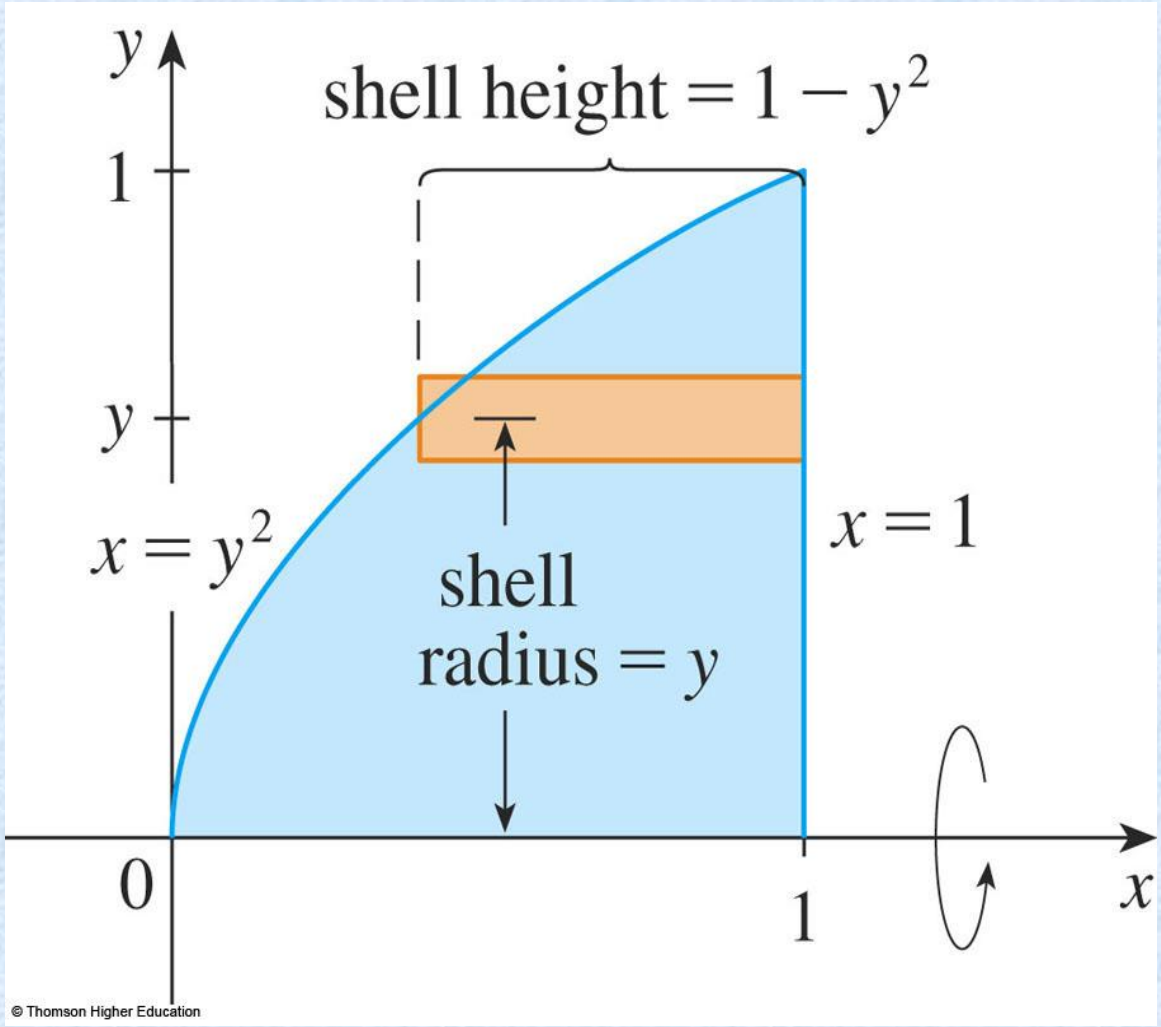


Example 3

Use cylindrical shells to find the volume of the solid obtained by rotating about the *x*-axis the region under the curve $y = \sqrt{x}$ from 0 to 1.

Remark:

This problem was solved using disks in Example 2 in Section 6.2


y
shell height = 1 − y²
1
y
x = y²
x = 1
shell
radius = y
0
1
x
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Example:

Find the volume of the solid obtained by rotating the region bounded by $y = x - x^2$ and $y = 0$ about the line $x = 2$.